

Air pollutants, genetic susceptibility, and incident schizophrenia in later life: A prospective study in the UK Biobank.

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Air pollution has been linked to multiple psychiatric disorders, but little is known on its long-term association with schizophrenia. The interaction between air pollution and genetic susceptibility on incident schizophrenia has never been reported. We aimed to explore the associations between long-term air pollution exposure and late-onset schizophrenia and evaluate whether genetic susceptibility could modify the association. This population-based prospective cohort study included 437,802 middle-aged and elderly individuals free of schizophrenia at baseline in the UK Biobank. Land use regression models were applied in the estimation of the annual average concentrations of nitrogen dioxide (NO₂), nitrogen oxides (NO_x), fine particulate matter (PM_{2.5}), and inhalable particulate matter (PM₁₀) at residence. The associations between air pollutants and schizophrenia were evaluated by using Cox proportional hazard models. A polygenic risk score of schizophrenia was constructed for exploring potential interaction of air pollutants with genetic susceptibility. An interquartile range increase in PM_{2.5}, PM₁₀, NO₂, and NO_x was associated with the hazard ratios (HR) for incident schizophrenia at 1.19, 1.16, 1.22, and 1.09, respectively. The exposure-response curves for the association of air pollution with incident schizophrenia were approximately linear. There are additive interactions of air pollution score (APS), PM₁₀, NO₂, and NO_x with genetic risk. Specifically, compared with participants with low genetic susceptibility and low APS, the HR was 3.23 for individuals with high genetic risk and high APS, among which 0.49 excess risk could be attributed to the additive interaction, accounting for 15 % of the schizophrenia risk. This large-scale, prospective cohort study conveys the first-hand evidence that long-term air pollution exposure could elevate schizophrenia incidence in later life, especially for individuals with higher genetic risks. The findings highlight the importance of improving air quality for preventing the late-onset schizophrenia in an aging era, especially among those with high genetic risks.

[Late-Onset Psychosis and Phenocopies].

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Psychotic symptoms can manifest at any age, but in the elderly they represent a real diagnostic challenge. Thought disorders, hallucinations (usually visual), mood disorders with delusions, impairment of social interaction and occasionally verbal or physical aggression may be observed (Karon & VandenBos, 1998). Since the first descriptions of classical psychiatry, attempts have been made to define the psychoses observed in the elderly and determine whether they are primary "psychiatric" syndromes or, conversely, whether they can be attributed to other pathologies. Thus, different concepts have emerged, such as Late Onset Psychosis or Late-Onset Schizophrenia, Very Late-Onset Psychosis or Very Late-Onset Schizophrenia-Like Psychosis – VLOSL, Late-Life Psychosis, etc.

Late-onset and nonlate-onset schizophrenia: A comparison of clinical characteristics in a multicenter study.

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Data are scarce regarding the potential clinical differences between non-late onset schizophrenia (NLOS,

i.e., disorder occurring before 40 years of age), late-onset schizophrenia (LOS, occurring between ages 40 and 60 years) and very-late-onset schizophrenia-like psychosis (VLOSLP, occurring after 60 years of age). Furthermore, previous research compared LOS patients with non-age matched NLOS patients. In this study, we sought to examine potential clinical differences between patients of similar age with LOS and NLOS. This is a cross-sectional multicentre study that recruited in- and outpatients older adults (aged ≥ 55 years) with an ICD-10 diagnosis of schizophrenia or schizoaffective disorder with NLOS and LOS. Sociodemographic and clinical characteristics, comorbidity, psychotropic medications, quality of life, functioning, and mental health care utilization were drawn for comparison. Two hundred seventy-two participants (79.8%) had NLOS, 61 (17.9%) LOS, and 8 (2.3%) VLOSLP. LOS was significantly and independently associated with greater severity of emotional withdrawal and lower severity of depression (all $p < 0.05$). However, the magnitude of these associations was modest, with significant adjusted odds ratios ranging from 0.71 to 1.24, and there were no significant between-group differences in other characteristics. In an age-matched multicenter sample of elderly patients with schizophrenia, older adults with LOS were largely similar to older adults with NLOS in terms of clinical characteristics. The few differences observed may be at least partially related to symptom fluctuation with time. Implications of these findings for pharmacological and nonpharmacological management is yet to be determined.

Utilizing structural MRI and unsupervised clustering to differentiate schizophrenia and Alzheimer's disease in late-onset psychosis.

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Late-onset psychosis (LOP) represents a highly heterogeneous and understudied condition, with potential origins ranging from atypically late onset of schizophrenia (SCZ) to Alzheimer's Disease (AD). Despite the clinical necessity of differentiating these conditions to guide effective treatment, achieving an accurate diagnosis remains challenging. This study aimed to utilize data-driven analyses of structural magnetic resonance imaging (MRI) to distinguish between these diagnostic possibilities. Utilizing publicly available datasets of MRI scans from 699 healthy control (HC) participants and 469 patients diagnosed with SCZ or AD, our analysis focused on bilateral subcortical volumetric measures in the caudate, hippocampus, putamen, and amygdala. We first trained an unsupervised K-means clustering algorithm based on SCZ and AD patients and achieved a clustering accuracy of 81 % and an area under curvature (AUC) of 0.79 in distinguishing between these two groups. Subsequently, we calculated the Euclidean distance between the AD and SCZ cluster centroids for each of ten patients with unexplained onset of psychosis after age 45 from a clinical MRI registry. Six patients were classified as AD and four as SCZ. Our findings revealed that among LOP participants, those classified in the SCZ cluster exhibited significantly greater right putamen volumes compared to those in the AD cluster ($p < 0.0025$). There were also intriguing clinical differences. While we do not have diagnostic biomarker information to confirm these classifications, this study sheds light on the heterogeneity of psychoses in late life and illustrates the potential use of widely available structural MRI and data-driven methods to enhance diagnostic accuracy and treatment outcomes for LOP patients.

Parenthood and all-cause mortality in older adults with schizophrenia: a multicenter 5-year prospective study.

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The large body of literature examining the association between parenthood and mortality in the general population contrasts with a lack of such studies on older adults with schizophrenia. Identifying potential protective factors of premature death in this population is important to help guide prevention measures. Here, we examined whether all-cause and cause-specific mortality rates significantly differ between older parents and non-parents with schizophrenia during a 5-year follow-up. We used data from a 5-year prospective multicenter sample of older adults with an ICD-10 diagnosis of schizophrenia (aged 55 years or more) recruited in France. We performed a forward stepwise logistic regression to examine the association between parenthood and all-cause mortality, including only independent variables that best explain outcome. Of the 323 older adults with schizophrenia, 133 (41.2%) were parents (mean age = 67.0, SD = 6.1) and 190 were not (mean age = 67.2, SD = 6.6). Following adjustments, parenthood was significantly associated with lower all-cause mortality compared to patients without children (21.1% [n=28] vs. 35.8% [n=68]; AOR = 0.50; 95%CI 0.27-0.94; p = 0.032); the association involved no significant sex differences. Parenthood could be a protective factor against mortality among older patients with schizophrenia who live in France. Further research is needed to understand the specific mechanisms underlying this association.

Very Late-Onset Schizophrenia-Like Psychosis: A Case Report and Critical Literature Review.

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Late-life psychosis presents a challenge, wherein a wide range of differential diagnoses should be considered. Very late-onset schizophrenia-like psychosis (VLOSLP) is a nosological entity that remains a conundrum. We provide a comprehensive literature review on the neurobiological underpinnings of VLOSLP. We describe a case that typifies the clinical presentation of VLOSLP. Although not pathognomonic, certain features, namely the two-stage progression of psychotic episode, partition delusions, multimodal hallucinations, and absent formal thought disorder or negative symptoms, are quite suggestive of VLOSLP. Several medical causes that might cause late-life psychosis, including neuroinflammatory/immunology diseases, were ruled out. Neuroimaging revealed basal ganglia lacunar infarctions along with chronic white matter small-vessel ischemic disease. The diagnosis of VLOSLP is based on clinical evidence, and the aforementioned clinical features support this diagnostic hypothesis. This case adds to the growing body of evidence pertaining to the relevance of cerebrovascular risk factors in the pathophysiology of VLOSLP, alongside age-specific neurobiological processes. We hypothesized that microvascular brain lesions disrupt the frontal-subcortical circuitry and uncover other core neuropathological processes. Future research should focus on identifying a specific biomarker that would allow clinicians to more accurately diagnose VLOSLP, differentiate it from other overlapping clinical entities such as dementia or post-stroke psychosis, and provide a tailored treatment for the patient.

Schizophrenia with onset after age 50 years. I: Phenomenology and risk factors.

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Schizophrenia occurring for the first time in late life may be a distinct entity or part of a continuum. Can late-onset schizophrenia (LOS) and early-onset schizophrenia (EOS) be differentiated by their phenomenology and risk factors to their development? Convenience samples of 27 DSM-III-R defined LOS

subjects, 30 EOS subjects and 34 control subjects were systematically assessed. Premorbidly, both groups of subjects with schizophrenia had personality traits that were different from controls but not from each other. The EOS subjects had more family members with a history of psychiatric illness or schizophrenia and less hearing impairment than the other two groups, which did not differ from each other. Clinically, LOS and EOS subjects were similar, except that EOS subjects had more negative symptom scores, tended to have more delusions of guilt/sin and of being controlled and more formal thought disorder, and had significantly poorer instrumental activities of daily living. Phenomenology and risk factors do not distinguish discrete LOS.

[Late schizophrenia (author's transl)].

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The designation "late schizophrenia" is used with reference to M. Bleuler. Of 502 cases of schizophrenia admitted to the mental hospital of the university of Bonn from 1945-1959 and later catamnesticly examined, 14% were taken ill after age 40. Including the deceased, the percentage increases to 17.1 (110 of 644 cases). The onset of disturbance could be ascertained with 50% occurring equally between age 40-44 and age 45-59. Late schizophrenia became manifest after age 50 in 24. 3% and in 3.1% after age 60. The average period of observation was 17.8 years, and 20.7 years including prodromes. In 42.8% prodromes and/or signs in advance could be traced. Females predominated (64.3%). Syntonic personalities were markedly more frequent (49.3%), psychopathic primary personalities less frequent than in other schizophrenias in the first half of life. With regard to premorbid intelligence or heritability no differences could be found. Psychic-reactive actuation (31.9%) was not significantly more frequent than in the group as a whole (25%). Not even by social factors (social class of parental family, bachelorhood rate, childlessness, and late marriage) could late schizophrenia be reliably distinguished from other schizophrenias. The quota of initially dominating paranoid-hallucinatory (55.4%) and paranoid syndromes (23.2%) is markedly higher than in schizophrenia of the first half of life, depressive moods (47.1%) and coenaesthesias (64.3%) less frequent. The final prognosis is - especially for women - more favorable, the rate of full remissions amounting 30%; 35% showed remissions to slight pure residual symptoms. All types of schizophrenic courses also occur in late schizophrenia; monophasic types are significantly more frequent, the two most prognostically unfavorable types are significantly rarer than in the group as a whole. The rate of social reintegration is with 64.3% higher and that of permanent hospitalization 10% lower than in the group as a whole. In view of the identity of late schizophrenia with all other schizophrenias, the same principles have to be applied to therapy and rehabilitation.

Clinical presentation of late-onset schizophrenia.

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DSM-III-R defines late-onset schizophrenia as schizophrenia that begins after age 45. The authors describe five patients who met DSM-III-R criteria for this disorder. The characteristic features of the disorder include bizarre delusions (usually of a persecutory nature), auditory hallucinations, a history of normal functioning at least until age 45, and a deterioration in personal-social functioning after the onset of illness. Physical examination, routine laboratory tests, neuropsychological evaluation, and magnetic resonance imaging (MRI) scans did not suggest the presence of any diagnosable organic mental disorder. In each of the five cases, mood disorders, delusional (paranoid) disorder, and schizophrenia with onset

before age 45 were also considered in the differential diagnosis and ruled out. The authors' case vignettes support the premise that late-onset schizophrenia is a clinically diagnosable entity. Cautions in the diagnosis and treatment of such patients are discussed.

Schizophrenia syndrome due to C9ORF72 mutation case report: a cautionary tale and role of hybrid brain imaging!

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Frontal variant frontotemporal dementia is a common cause of presenile dementia. A hexanucleotide expansion on chromosome 9 has recently been recognized as the most common genetic mutation cause of this illness. This sub-type tends to present psychiatrically with psychosis being a common presenting symptom before the onset of cognitive changes or brain atrophy. A few case series have been published describing the prominence of early psychotic symptoms, and lack of clear brain atrophy on clinical brain imaging imposing a challenge in reaching early accurate diagnosis. In this report, we present a case whereby the diagnosis of Schizophrenia syndrome was made and the patient was treated for years with multiple interventions for that syndrome before reaching the accurate diagnosis of Frontal variant frontotemporal dementia due to hexanucleotide expansion on chromosome 9. This diagnosis was confirmed after genetic testing and findings on a hybrid Positron Emission Tomography/Magnetic Resonance Imaging scanning. A 60-year-old female diagnosed with schizophrenia at age 50 after presenting with delusions and hallucinations, which proved to be refractor to several lines of pharmacological and non-pharmacological interventions including electroconvulsive therapy. Patient had a history of post-partum psychosis in her 20s. She was referred to cognitive neurology due to progressive decline in function. While clinical structural brain imaging data were not adequate to support an alternative neurological diagnosis, careful inquiry elicited a history of psychotic illness followed by progressive decline in a sister. Genetic testing confirmed hexanucleotide expansion on chromosome 9 mutation. The patient was offered a state-of-the-art FD-Glucose Positron Emission Tomography/Magnetic Resonance Imaging scan available at our centre. While volumetric Magnetic Resonance Imaging scan did not show volume loss in frontotemporal areas, the hybrid scan showed regionally specific deficit in FD-Glucose Positron Emission Tomography affecting medial superior frontal, insula, inferior temporal, thalamus, and anterior cingulate cortex consistent with behavioral variant frontotemporal dementia. This case highlights the importance of considering Frontal variant frontotemporal dementia due to hexanucleotide expansion on chromosome 9 when facing relatively late-onset, refractory schizophrenia-like syndrome. Careful history from all available sources to elicit family history of similar presentation is very important. Genetic testing and functional brain imaging can aid in confirming the diagnosis and potentially streamlining the management of these cases.